

ABHISHEK PANDIT

Senior Avionics Software Engineer | Safety-Critical Embedded Systems | DO-178B/C | RTOS

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PROFESSIONAL SUMMARY

Avionics software engineer with 8+ years of experience across the full embedded software lifecycle — requirement engineering, static and dynamic architecture, high/low level design, coding, and unit verification — for safety-critical FADEC and avionics ECU platforms on civil aerospace programs, developed to DO-178C Level A/B objectives. Hands-on in Embedded C/C++ and RTOS-based real-time design (FreeRTOS, bare-metal, ARM Cortex/STM32), with bidirectional requirements traceability (IBM DOORS), MC/DC structural coverage, and configuration/change management practices aligned to DO-178 and ARP 4754A/4761. Experienced coordinating with systems, hardware, test, and certification stakeholders to resolve complex technical issues, using communication protocols (CAN, SPI, UART, I2C, ARINC) and V&V; tooling (VectorCAST, Polyspace).

CORE COMPETENCIES

- Languages & RTOS: Embedded C, C++, Python | FreeRTOS, Bare-Metal | MATLAB/Simulink Model-Based Design
- Requirements & Design Engineering: Requirement gathering & analysis, static/dynamic software architecture, high-level & low-level design, incremental design evolution, IBM DOORS traceability
- Platforms & Architecture: STM32, ESP32, ARM Cortex microcontrollers | Hardware-software integration
- Protocols: CAN, SPI, UART, I2C, ARINC, MQTT, LoRa, BLE
- Certification & Standards: DO-178C (Level A/B), ARP 4754A, ARP 4761, DO-160, AS9100, CMMI Awareness
- V&V; & Debug Tools: VectorCAST, Polyspace Code Prover, MC/DC structural coverage, HIL/SIL, JTAG, oscilloscope & logic analyzer debugging, IBM DOORS traceability
- Practices & Tools: Agile/Scrum, Git, JIRA, code & design reviews, configuration management

PROFESSIONAL EXPERIENCE

Senior Designer – Embedded Software, Engine Control Systems — Mar 2024 – Present
Tata Consultancy Services (TCS) | Bengaluru, India

Designed and developed real-time embedded software for FADEC and avionics ECU platforms across the full lifecycle — requirements, design, integration, and system-level validation — on a civil propulsion OEM program, following DO-178C Level A/B and aerospace industry standards.

Led requirement engineering activities — gathering system requirements and analyzing/documenting software requirements — and defined system interfaces in coordination with systems, hardware, and controls teams, establishing bidirectional requirements traceability in IBM DOORS from system requirements through V&V; evidence.

Performed debugging, optimization, and validation of embedded control software, achieving 100% MC/DC structural coverage on safety-critical modules via VectorCAST unit testing and Polyspace static analysis.

Built reusable verification checklists for MATLAB/Simulink model-based design reviews and auto-generated code validation, cutting review turnaround time by ~30%.

Led high-level and low-level design reviews, code reviews, and documentation prepared to DO-178C objectives, coordinating with verification teams and certification stakeholders to resolve technical issues and support on-schedule closure.

Senior Engineer – Embedded Software & Verification — May 2022 – Mar 2024
L&T Technology Services | Bengaluru, India

Developed and integrated embedded software for ECU platforms in Embedded C using AUTOSAR and bare-metal architectures, supporting SIL/HIL test campaigns that validated control algorithms against system-level requirements.

Built and debugged CAN, SPI, and UART drivers, using JTAG and bench instrumentation to isolate and resolve hardware-software integration issues during bring-up and test.

Redesigned diagnostic routines and added FMEA-driven test cases for a redundancy management module, achieving zero safety-critical escapes during independent verification review.

Mentored junior engineers on embedded driver development and verification workflows, contributing to a ~25% reduction in defect leakage across two program cycles.

Participated in design, code, and readiness reviews, maintaining controlled documentation per aerospace configuration management practices.

IoT & Embedded Engineer — Jul 2018 – May 2022
AhamCo / Bliss Connect / Accentus Consulting Pvt. Ltd. | Kolkata, India

Designed and deployed firmware on STM32, ESP32, and ARM Cortex microcontrollers for smart home and industrial automation products, including MQTT/AWS IoT cloud integration.

Built custom PCB layouts (Altium/KiCad) and firmware (C/C++, Python) for sensor nodes with LoRa and BLE connectivity, enabling low-power field deployments with 6–12 month battery life.

Established firmware development and testing practices across small product teams, shortening concept-to-prototype cycles by an average of 3 weeks per release.

CERTIFICATIONS & TRAINING

- Mastering RTOS: Hands-on FreeRTOS and STM32Fx with Debugging
- Mastering Microcontroller: Timers, PWM, CAN, Low Power (MCU2)
- Mastering Microcontroller and Embedded Driver Development (I2C, SPI, USART)
- Exploring AWS IoT

EDUCATION

Bachelor of Engineering – Electronics & Communications — 2014 – 2018
Narula Institute of Technology (NIT), West Bengal